MIL-P-44206B
13 October 1988
SUPERSEDING
MIL-P-44206A
18 April 1986

MILITARY SPECIFICATION

POTATOES AU GRATIN, THERMOSTABILIZED, FOR MEAL, READY-TO-EAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers thermostabilized potatoes au gratin in flexible packages for use as a component of the Meal, Ready-to-Eat, Individual.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-44073 - Packaging and Thermoprocessing of Foods in Flexible Pouches

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks, and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces.

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies may be obtained from the Office of Drinking Water, Environmental Protection Agency, 401 M Street, S.W., Washington, DC 20406.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Grades of Cheddar Cheese

Dairy Plants Surveyed and Approved for USDA Grading Service

(Copies may be obtained from the Dairy Standardization Section, Dairy Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2750, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

U.S. Standards of Identity for Margarine

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

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(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Copies should be obtained from the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Copies should be obtained from the American Dehydrated Onion and Garlic Association, 650 California Street, Suite 800, San Francisco, CA 94108.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Copies should be obtained from the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Copies should be obtained from the National Academy Press, 2101 Constitution Avenue, Washington, DC 20418.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.2).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-flavors, off-odors, and off-colors.

3.2.1 Potatoes, diced, dehydro-frozen. The dehydro-frozen potatoes shall be firm, have a good color and flavor, and be of a white flesh variety suitable for canning. The maximum specific gravity for the potatoes shall be 1.075 with a reducing sugar content of not more than 2.0 percent on a dry weight basis. The potatoes shall be mechanically diced to approximately 3/8 inch cubes.

3.2.2 Cheese, cheddar, sharp, dehydrated, uncolored. Dehydrated cheddar cheese shall be derived from aged cheddar cheese. The dehydrated cheese shall be uncolored, uniform in composition and appearance, and free from burnt specks and from lumps that do not fall apart under light finger pressure. The cheese shall possess a characteristic flavor and odor. The dehydrated cheese shall contain not more than 4.0 percent moisture, not more than 5.0 percent salt, and not less than 50.0 percent milkfat (see 6.3).

3.2.3 Margarine. Margarine shall comply with the FDA Standard of Identity for Margarine and shall possess a fine and pleasing flavor. The body shall be smooth, firm, and homogeneous. The margarine shall possess a uniform yellow color and may be salted or unsalted.

3.2.4 Shortening, vegetable, powdered. Powdered vegetable shortening shall consist of components which, when combined, will serve as a product whitening agent. The powdered shortening, when used as an ingredient in this product, shall withstand the specified thermoprocessing conditions without browning. A typical analysis of the powdered shortening follows:

Fat	75.00 (± 1.50) percent
Protein	5.50 (± 0.50) percent
Carbohydrates	15.00 (± 1.00) percent
Moisture	2.00 percent maximum

The powdered vegetable shortening shall comply with MIL-STD-900.

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3.2.5 Salt. Salt shall be noniodized, white, refined sodium chloride, with or without anticaking agents, and shall comply with purity standards for sodium chloride of the Food Chemicals Codex.

3.2.6 Starch, waxy maize, modified. Starch shall be white, odorless, finely pulverized, modified waxy maize starch for use in thermostabilized foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140°F to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress, and under conditions of cold storage. The starch shall be bland with essentially no cereal or starch taste and shall comply with MIL-STD-900.

3.2.7 Onion powder. Onion powder shall be Fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.8 Pepper, white, ground. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or the outer and inner coverings have been removed. The ground white pepper shall have a characteristic penetrating odor, a hot biting pungent flavor, and a light color. The white pepper shall contain not less than 1.0 mL of volatile oil per 100 grams of ground white pepper and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

3.2.9 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formulation.

3.2.10 Water. Water used for formulation, rehydration, blanching, ice-making, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.11 Color, Annatto, dry. Annatto color shall be a reddish-orange water soluble powder of the extractive of Annatto bixa Orellana. The powder shall contain not more than 5.0 percent moisture and 100 percent of the powder shall pass through a U.S..Standard No. 10 sieve. The total color (as percent Norbixin) shall be 15 percent.

3.3 Preparation and further processing. Preparation and further processing shall be on a continuous basis.

3.3.1 Preparation of dehydro-frozen potatoes. The dehydro-frozen potato dices shall be blanched to approximately 1.5 times the frozen weight. The blanched potatoes shall be handled in such a manner as to prevent discoloration and shall be filled into the pouches within 4 hours after blanching.

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3.3.2 Sauce formulation and preparation. The sauce shall be formulated and prepared as follows:

Ingredients	Percent by weight
Water	80.963
Cheese, cheddar, dehydrated	8.24
Margarine	4.12
Shortening, vegetable, powdered	3.64
Starch, waxy maize, modified	2.11
Salt 1/	0.52
Onion powder	0.34
Pepper, white	0.06
Annatto	0.007

1/ The total amount of salt in the sauce formula shall be adjusted as necessary to produce a product that complies with the finished product salt requirements (see 3.6).

NOTE: The following sauce preparation procedures were used in the development of this product. Alternate procedures may be used provided finished product requirements are met. (When alternate procedures are used, the time and temperature requirements specified for the prepared sauce are still applicable.)

a. A slurry shall be made from the starch and part of the water.

b. The remaining ingredients shall be heated in a steam jacketed kettle to a temperature of 180° to 190°F.

c. The starch slurry shall be added and stirred vigorously and continuously until the sauce reaches a temperature of 180° to 190°F.

d. The volume of the final mixture shall be adjusted with water to compensate for evaporation loss during heating and holding.

e. The sauce shall be maintained in the temperature range of 150° to 180°F and filled within 4 hours after preparation.

3.4 Pouch filling and sealing. Each pouch (see 5.1) shall be filled with product so as to conform to the finished product requirements and to the following requirements:

a. Each pouch shall be filled and sealed in accordance with the pouch filling and sealing requirements specified for class 2 of MIL-P-44073.

b. Each filled and sealed pouch shall be in the retort process within 2 hours after sealing.

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3.5 Pouch thermoprocessing. The filled and sealed pouches shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved. The thermoprocessing operation shall be in compliance with the thermoprocessing requirements for class 2 of MIL-P-44073.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insects, insect parts, hair, wood, paper, paint, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no color foreign to the product.

d. The average net weight shall be not less than 5.0 ounces.

e. No individual pouch shall contain less than 4.5 ounces of product.

f. The average drained weight shall be not less than 2.2 ounces or more than 2.8 ounces.

g. No individual pouch shall contain less than 2.0 ounces of drained potato dice.

h. The texture of the potato dice shall not be mushy, hard, fibrous, or tough.

i. The potato dice shall be approximately uniform in size and shape.

j. The sauce consistency shall be smooth, without lumps, and shall not be excessively thick or thin.

k. The salt content of the product in any individual pouch shall be not less than 0.5 percent nor greater than 1.0 percent.

l. The product shall show no evidence of excessive heating (materially darkened or scorched).

3.6.1 Palatability. The finished product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods", and the plant sanitation requirements of the appropriate Government inspection agency.

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3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

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4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved products.

4.5.3 Filled and sealed pouch inspection. Inspection of filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44073.

4.5.4 Net weight inspection. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 4.5 ounces shall be scored as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 0.1 ounce. In addition, the lot shall be rejected if the sample average net weight is less than 5.0 ounces.

4.5.5 Product and drained weight inspection.

4.5.5.1 Product inspection. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature (65° to 75°F) after completion of the thermoprocessing operation. The pouches shall be heated in 150° to 160°F water for 10 minutes, opened, and inspected. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The sample size shall be as indicated by the sampling plan specified in table I. The contents shall be inspected for the defects indicated in table II.

TABLE I. Double sampling plan for product inspection 1/

Lot size (pouches)	Sample size (pouches)	Cumulative sample	Acceptance number	Rejection number
0 to 500,000	20	--	0	2
	20	40	1	2

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

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c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second sample shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product defects 1/ 2/

Category	Defect
Major	
101	Potato texture mushy, hard, fibrous, or tough.
102	Potato dice not approximately uniform in size and shape.
103	Sauce consistency lumpy or is excessively thick or thin.
104	Product shows evidence of excessive heating (materially darkened or scorched).

1/ The presence of foreign material (e.g., glass, dirt, insects, insect parts, hair, paper, paint, wood, metal), foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, musty, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

4.5.5.2 Drained weight inspection. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature (65° to 75°F) after completion of the thermoprocessing operation. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The sample size shall be as indicated by the double sampling plan specified in table III. The drained weight shall be determined and inspected for the defect listed in table IV.

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TABLE III. Double sampling plan for drained weight inspection 1/ 2/

Lot size (pouches)	Sample size (pouches)	Cumulative sample	Acceptance number	Rejection number
0 to 500,000	20	--	0	2
.	20	40	1	2

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second sample shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

2/ To determine drained weight, the pouch contents shall be poured into a flat bottom container. A minimum of three times the pouch's volume of 180° to 190°F water shall be added to the container so as to cover the contents. The contents and water shall be gently agitated such as to separate sauce from potatoes. The contents shall then be poured into a U.S. Standard 1/4 inch sieve in a manner that will distribute the product over the sieve. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at an approximate 45 degree angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

TABLE IV. Drained weight defect

Category	Defect
Major	
101	Drained weight of diced potatoes less than 2.0 ounces. 1/

1/ If the sample average drained weight is less than 2.2 ounces or more than 2.8 ounces, the lot shall be rejected.

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4.5.6 Salt content testing. The unopened sample pouches shall be warmed in a water bath to melt fat adhering to the inside of the pouches. The sample pouches shall be opened and the entire contents of each pouch shall be separately blended in a Waring blender or equivalent. The salt content of the product from each pouch in the sample shall be determined in accordance with the Official Methods of Analyses of the Association of Official Analytical Chemists, Chapter: Meat and Meat Products. Test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the salt content requirement in 3.6 shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.7 Packaging inspection. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-P-44073.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the requirements of MIL-P-44073 .

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document
b. When a first article is required (see 3.1, 4.4, and 6.2).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Cheese, cheddar, sharp, dehydrated. Kraft Cheeztang, spray dried, uncolored, product No. 9378 and Land O Lakes spray dried, white, product No. 49122 were found to be satisfactory in the development of this product.

6.4 Subject term (key word) listing.

Potatoes
Meal, Ready-To-Eat
Ration
Thermostabilized

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, TS
Navy - MC
DP - SS

Preparing activity:

Army - GL

Project No. 8940-0651

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-44206B	Potatoes Au Gratin, Thermostabilized, For Meal, Ready-To-Eat

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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